

# The residual bound, step by step

figures `fig_kernel_free.png`, `fig_design_range.png`; scripts listed in the Reproduction section

## Symbols

|                           |                                                                                                  |                            |                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------|
| $\tau, \tau_{\text{end}}$ | time since onset; window end                                                                     | $x$                        | $= C_2 \tau_{\text{end}}$ , dimensionless window           |
| $y_i$                     | measured rate at $\tau_i$                                                                        | $N$                        | number of samples in the window                            |
| $C_1, C_2$                | nominal amplitude $K\dot{M}$ and exponent                                                        | $K, \dot{M}$               | calibrated gain; ramp rate                                 |
| $W, z_{CoM}, J_P$         | weight, CoM height, pivot inertia                                                                | $W z_{CoM}$                | $= J_P C_2^2$ (the identity)                               |
| $e_\omega(\tau)$          | true rate minus the nominal                                                                      | $b, n$                     | gyro bias; disturbance                                     |
| $C$                       | fitted baseline (pre-onset median)                                                               | $r_i$                      | residual $y_i - \hat{\omega}_i$                            |
| $\delta t_c, \beta$       | onset shift of the witness branch; its<br>sinh coefficient, $\tanh(C_2 \delta t_c) = -\beta/C_1$ | $\delta e_\omega$          | what the shift cannot absorb                               |
| $\rho(s)$                 | deviation forcing (small-angle + GE)                                                             | $\bar{\rho}$               | window average bound on $\rho$ , eq. (91)                  |
| $\dot{\rho}$              | its time derivative                                                                              | $\varphi_{\text{max}}$     | 10° design box                                             |
| $\omega_{\text{max}}$     | nominal peak rate $K\dot{M} \sinh x$                                                             | $l_{\text{arm}}, \beta_M$  | pivot arm; GE coefficient                                  |
| $\Delta M_{\text{win}}$   | in-window moment excursion                                                                       | $E(\tau)$                  | envelope $\bar{\rho} \sinh(C_2 \tau)/(J_P C_2)$ , eq. (90) |
| $n_{\text{hi}}$           | residual content above 5 Hz                                                                      | $\kappa_q, s_{\text{med}}$ | quiet-window shape ratio; median                           |
| $\hat{n}$                 | noise estimate, Sec. 6                                                                           |                            | in-window enhancement (1.60)                               |

## 1 The measurement and the fit

The record in the window and the fitted model are

$$y_i = C_1 (\cosh(C_2 \tau_i) - 1) + e_\omega(\tau_i) + b + n_i, \quad (1)$$

$$\hat{\omega}_i = C_1^* (\cosh(C_2^* (\tau_i - \delta t_c^*)) - 1) + C, \quad (2)$$

and on the pre-onset segment the baseline matches the bias,

$$C \approx b. \quad (3)$$

## 2 The residual, and what the shift absorbs

Subtracting (2) from (1) with (3),

$$r_i = y_i - \hat{\omega}_i = \underbrace{C_1 (\cosh(C_2 \tau_i) - 1) + e_\omega(\tau_i)}_{\text{to be absorbed}} - C_1^* (\cosh(C_2^* (\tau_i - \delta t_c^*)) - 1) + n_i. \quad (4)$$

The question is how much of the braced part a member of the family can match. Why a sinh term, of all shapes? Two facts meet. First,  $e_\omega$  *grows* like sinh: its envelope (90) is  $E(\tau) \propto \sinh(C_2 \tau)$ , because the deviation rides the instability's own growing mode — so peeling off a sinh multiple targets the dominant part of the error. Second, sinh is exactly what the family can absorb: it is the tangent direction of the onset shift,  $\partial_{\delta t_c} C_1 (\cosh(C_2 (\tau - \delta t_c)) - 1)|_0 = -C_1 C_2 \sinh(C_2 \tau)$ , so adding  $\beta \sinh$  lands inside the family (made exact in (6)–(7)). Had the envelope been, say,  $\tau^2$ -shaped, subtracting  $\beta \tau^2$  would target the error just as well and help not at all — the family could not represent it. The coincidence of the two — the error's dominant shape *is* the direction the fit can absorb — is the entire reason the measured residuals sit flat. Add and subtract that one sinh term — this *defines*  $\delta e_\omega$ , and the equality holds at every  $\tau$ , both window ends included:

$$C_1 (\cosh(C_2 \tau) - 1) + e_\omega(\tau) = \underbrace{C_1 (\cosh(C_2 \tau) - 1) + \beta \sinh(C_2 \tau)}_{\hat{h}(\tau), \text{ the witness}} + \delta e_\omega(\tau). \quad (5)$$

The witness is a member of the family in disguise. With the addition identity

$$\cosh(C_2 (\tau - \delta t_c)) = \cosh(C_2 \delta t_c) \cosh(C_2 \tau) - \sinh(C_2 \delta t_c) \sinh(C_2 \tau), \quad (6)$$

matching the cosh and sinh coefficients  $[R \cosh(C_2 \delta t_c) = C_1, -R \sinh(C_2 \delta t_c) = \beta]$  gives, *exactly*,

$$\hat{h}(\tau) = R(\cosh(C_2(\tau - \delta t_c)) - 1) + (R - C_1), \quad R = \sqrt{C_1^2 - \beta^2}, \quad \tanh(C_2 \delta t_c) = -\beta/C_1, \quad (7)$$

a branch with amplitude  $R$ , onset  $\delta t_c$ , and a baseline offset  $R - C_1$  — three numbers, all inside the family’s freedoms in (2) (the offset is absorbed by  $C$ : the witness’s baseline slot carries  $b + (R - C_1)$ , not  $b$  alone, and the “ $\approx$ ” of (3) is exactly this second-order constant; what it costs, and on which segment it lands, is settled in the remark closing the pre-onset section). On the campaign  $\beta/C_1 \leq 0.023$ , so the amplitude and baseline adjustments are second order ( $|R/C_1 - 1| \leq 0.03\%$ ); the shift carries the first-order content. The remainder is

$$\delta e_\omega(\tau) = e_\omega(\tau) - \beta \sinh(C_2 \tau). \quad (8)$$

### 3 Choosing the shift: match the true curve at the window end

One free number,  $\delta t_c$  (equivalently  $\beta$ ), and one choice: make the shifted branch meet the true curve at  $\tau_{\text{end}}$ . Setting  $\delta e_\omega(\tau_{\text{end}}) = 0$  in (8),

$$\beta = \frac{e_\omega(\tau_{\text{end}})}{\sinh(C_2 \tau_{\text{end}})}, \quad \delta t_c = -\frac{1}{C_2} \operatorname{artanh}\left(\frac{\beta}{C_1}\right) \approx -\frac{e_\omega(\tau_{\text{end}})}{C_1 C_2 \sinh(C_2 \tau_{\text{end}})}. \quad (9)$$

Since  $e_\omega > 0$  (the true curve rises faster than the nominal),  $\delta t_c < 0$ : the branch starts *earlier* than the true onset. On the campaign  $\beta/C_1 = 0.010\text{--}0.023$ , i.e.  $|\delta t_c| = 2\text{--}4$  ms.

*Admissibility — why  $|\beta| < C_1$  is guaranteed.* (7) needs  $R$  real, i.e.  $|\beta| < C_1$ ; otherwise the witness is not a cosh branch at all. This is provable, not assumed: the envelope (90) gives  $|e_\omega(\tau_{\text{end}})| \leq \bar{\rho} \sinh(C_2 \tau_{\text{end}})/(J_P C_2)$ , so  $|\beta| \leq \bar{\rho}/(J_P C_2)$ , and dividing by  $C_1 = \dot{M}/W z_{CoM}$  with the identity  $J_P C_2^2 = W z_{CoM}$ ,

$$\frac{|\beta|}{C_1} \leq \frac{\bar{\rho} C_2}{\dot{M}} \triangleq u. \quad (9a)$$

So  $|\beta| < C_1$  holds whenever  $\bar{\rho} < \dot{M}/C_2$  — *the ramp must sweep more moment in one e-folding time  $1/C_2$  of the instability than the window-averaged remainder*. A protocol that violated this would have its excitation comparable to the model error it is meant to dominate, so any meaningful design satisfies it with room:  $u \leq 0.59$  at the campaign’s slowest ramp,  $\leq 0.66$  at the worst design-box corner of Sec. 9, and the realised  $\beta/C_1 = 0.010\text{--}0.023$  sits thirty times below the ceiling. The same  $u$  is the argument of the  $\operatorname{artanh}(u)/u$  factor in (21): one dimensionless number controls both the witness’s admissibility and the exactness of the shift ceiling.

*What if  $u \geq 1$  — ramps slower than  $\bar{\rho} C_2$ .* Three things, in decreasing order of certainty. (i) Only the *a priori* guarantee is lost: (9a) is sufficient, not necessary, so the witness may well still be admissible — the realised  $\beta/C_1$  sits some thirty times below the envelope ceiling on the campaign — but nothing proves it before the experiment any more. (ii) If  $|\beta| \geq C_1$  did occur, the combination  $C_1 \cosh(C_2 \tau) + \beta \sinh(C_2 \tau)$  is no longer a cosh branch: it equals  $\sqrt{\beta^2 - C_1^2} \sinh(C_2 \tau + \varphi)$  (a pure exponential at  $|\beta| = C_1$ ), and no real time shift of the family (2) draws a sinh-type curve. Step (10) then has no admissible competitor to hand to the minimiser, and the cap loses its proof — even though the Dirichlet bound (11)–(17) on  $e_\omega - \beta \sinh(C_2 \tau)$  itself survives, since it never used admissibility. Moving the matching point earlier does not rescue it: the envelope of  $e_\omega$  and  $\sinh(C_2 \tau)$  grow in the same ratio, so the required  $\beta$  ceiling is the same at every matching point. (iii) Physically,  $\dot{M} < \bar{\rho} C_2$  says the moment swept in one e-folding of the instability no longer dominates the model remainder — the identification concept itself, not just its certificate, is degrading; the campaign’s slowest ramp already runs at  $u = 0.59$ , close to that boundary. Numerically,  $u = 1$  is reached at  $\dot{M} = \bar{\rho} C_2 = 0.059$  N m/s for the campaign vehicle and at  $\dot{M} = 0.054$  N m/s at the worst design-box corner of Sec. 9 ( $m = 3.22$  kg,  $l_p = 0.140$  m,  $p_{\text{off}} = -0.020$  m,  $z_{CoM} = 0.20$  m); a hypothetical  $\dot{M} = 0.03$  N m/s ramp at that corner would have  $u = 1.81$ . So (9a) doubles as an analytical *lower* rate limit for the protocol,  $\dot{M} > \bar{\rho} C_2 \approx 0.06$  N m/s — about half the slowest campaign rate — complementing the allocator-saturation *upper* limit on  $\dot{M}$ . In practice the logic runs forward, and this is the intended reading: the guarantee fails *only* there, and since  $\bar{\rho}$  and  $C_2$  need nothing but the coarse design-box ranges known before any experiment, one computes the box-worst  $\bar{\rho} C_2 = 0.054$  N m/s in closed form first and then schedules the ramp rates above that floor with margin — the campaign’s  $\dot{M} \geq 0.10$  N m/s is a factor  $\approx 2$  above it, giving  $u \leq 0.66$  everywhere in the box.

*Could the family be composed so the problem never arises?* Partly, and where it splits is instructive. Replace the shifted-cosh family (2) by its linear span, fitting  $\hat{\omega} = A(\cosh C_2 \tau - 1) + B \sinh C_2 \tau + C$ : the witness  $\omega_{\text{nom}} + \beta \sinh(C_2 \tau)$  is then a member for *every*  $\beta$  (take  $A = C_1$ ,  $B = \beta$ ,  $C = b$ ), so the argmin step (10)

holds unconditionally and admissibility disappears from the residual cap altogether. The added column is no stranger:  $\sinh$  is exactly the tangent direction of the time shift ( $\partial_{\delta t_c}$  of the branch at  $\delta t_c = 0$  is  $-C_1 C_2 \sinh$ , Appendix A.2), so this is the linearised version of the same estimator. The cost is one more degree of freedom in the noise term (minor) and a near-collinear pair of columns —  $\cosh - 1$  and  $\sinh$  agree to  $O(e^{-2x})$  at the window end — so the residual stays clean while the individual coefficients  $A, B$  get noisy: the ridge of Sec. 4, resurfacing as conditioning. What *cannot* be designed away is the readout. Converting  $(A, B)$  back into an onset time needs  $R = \sqrt{A^2 - B^2}$  real, i.e.  $|B| < A$  — the very same condition, now appearing where it belongs.  $u < 1$  is not an artifact of the cosh parametrisation; it is the statement that the data actually describes a delayed cosh onset rather than a sinh-type curve, and no reparametrisation makes data interpretable that is not. The practical split: the linear family buys an *unconditional* residual cap, and  $u < 1$  then governs only whether the fitted curve has a physical onset to read off.

Then, because the minimiser in (2) is at least as good as this one particular shifted branch,

$$\boxed{\sum_{i=1}^N r_i^2 \leq \sum_{i=1}^N (\delta e_\omega(\tau_i) + n_i)^2 \implies \text{RMS}(r) \leq \text{RMS}(\delta e_\omega) + \text{RMS}(n).} \quad (10)$$

Everything now rests on one question: **how big is  $\delta e_\omega$ ?**

## 4 In exactly what sense the true curve is a time-shifted nominal

The ansatz of (5) deserves a precise statement, because “the true model is the nominal, time-shifted” is true in one sense, exactly, and false in another, and the gap between the two *is*  $\delta e_\omega$ .

*The local (osculating) sense — exact at every instant.* Splitting the manuscript’s Duhamel kernel (89) with the addition identity gives the deviation two running coefficients,

$$e_\omega(\tau) = A(\tau) \cosh(C_2 \tau) - B(\tau) \sinh(C_2 \tau), \quad A = \frac{1}{J_P} \int_0^\tau \cosh(C_2 s) \rho ds, \quad B = \frac{1}{J_P} \int_0^\tau \sinh(C_2 s) \rho ds, \quad (10a)$$

so the true structural curve is  $(C_1 + A) \cosh(C_2 \tau) - B \sinh(C_2 \tau) - C_1$ . Freezing the coefficients at an instant  $\tau$  and applying the shift identity (6) backwards, the curve agrees there — in value *and* slope — with exactly one branch, the *osculating branch*, whose parameters are

$$R(\tau) = \sqrt{(C_1 + A(\tau))^2 - B(\tau)^2}, \quad \tanh(C_2 \delta t_{\text{loc}}(\tau)) = \frac{B(\tau)}{C_1 + A(\tau)}. \quad (10b)$$

At the onset  $A = B = 0$ , so  $R = C_1$  and  $\delta t_{\text{loc}} = 0$ : **the true curve starts as the nominal, exactly.**

As the window progresses the osculating parameters drift monotonically (measured on the exact tip-over:  $\delta t_{\text{loc}} = 0$  at the onset, still only 2.5/0.4/0.1 ms at mid-window for  $\dot{M} = 0.10/0.45/1.20$ , then 546/239/85 ms at the window end, with  $R(\tau_{\text{end}})/C_1 = 13.0/3.1/1.5$ ). The late blow-up is structural:  $e_\omega$ ’s tail grows like  $e^{2C_2 \tau}$ , faster than any branch, so the branch that hugs it must run away in amplitude and shift together — and those two parameters are ridge-degenerate, wildly different  $(R, \delta t)$  pairs drawing nearly the same curve over the window. This is the same degeneracy that makes  $(C_1^*, C_2^*, \delta t_c^*)$  unidentifiable from the residual.

*The window ( $L^2$ ) sense — the estimator’s sense.* No single branch can follow the drifting profile (10b); a fit must pick one point of it. The least-squares stationarity  $\langle r, \sinh \rangle = 0$  weights the choice by  $\sinh^2(C_2 \tau)$ , which concentrates 63% of its mass in the last e-folding  $1/(2C_2) \approx 90\text{--}140$  ms of the window (86% in the last two), so the fitted single shift is essentially the end-matching one of (9) — and it is *small*:  $\beta/C_1 = 1.0\text{--}2.3\%$ , i.e.  $|\delta t_c| = 2\text{--}4$  ms on the campaign. In this sense the true curve *is* a slightly-shifted nominal: one branch, shifted by milliseconds, matches it over the whole window up to exactly  $\delta e_\omega$  — which vanishes at both ends and sags mid-window where the drifting profile is farthest from its end value. That sag is what (11)–(17) bound.

One sign remark connects the two senses. The osculating description carries a *larger* amplitude with a *later* onset ( $\delta t_{\text{loc}} > 0$ ); the witness of (9) carries the nominal amplitude with an *earlier* onset ( $\delta t_c < 0$ ). Along the ridge these describe nearly the same curve — larger-and-later  $\approx$  same-and-earlier — and the deployed estimator, whose amplitude is pinned to  $K\dot{M}$ , realises the earlier-onset representative: the sign observed on the campaign, and the sign whose  $M_{\text{crit}}$  cost (21) charges.

*Remark — a perfect fit at all  $N$  samples is impossible, and that is not a defect of the fit.* Two reasons, one counting and one dynamical. Counting: a branch has four free numbers  $(C_1^*, C_2^*, \delta t_c^*, C)$ , so it can be made to pass through at most four chosen samples; the other  $N - 4$  dimensions of the sample space are out of its

reach, and  $e_\omega$  has components there. Dynamical, and stronger: if a branch agreed with the true curve on any subinterval, then  $\delta e_\omega \equiv 0$  there, and (13) would force  $\dot{\rho} \equiv 0$  on that subinterval — the true curve solves the *forced* equation, the branches solve the homogeneous one, and they can coincide on an interval only where the forcing vanishes. Exact agreement at  $\tau_1, \dots, \tau_N$  is therefore not “difficult”; for  $\dot{\rho} \not\equiv 0$  it is unattainable in principle.

This is why the section’s claim is framed as it is. “The true curve is a time-shifted nominal” cannot be asserted exactly; asserted with the remainder, it *cannot be wrong*: eq. (5) is an identity —  $\delta e_\omega$  is defined as whatever the shifted branch misses — and the entire scientific content lives in one place, the bound (17) on that remainder. The structure is that of Taylor’s theorem: “ $f$  is approximately a polynomial” is a hope, “ $f$  equals the polynomial plus a remainder obeying a stated bound” is a theorem. Here the remainder is  $0.16\text{--}0.17^\circ/\text{s}$  realised, capped by (17) at  $0.5\text{--}0.9^\circ/\text{s}$  from pre-experiment constants, and located — zero at both ends, sagging mid-window — so the imperfection of the time-shift description is quantified in size, ceiling and shape.

The same statement has a clean linear-algebra reading: the model family is a four-parameter manifold in the  $N$ -dimensional sample space, the true record carries a component transverse to it that no choice of parameters can reach, and  $\delta e_\omega$  is exactly that transverse coordinate — introduced so that the description becomes an identity rather than an approximation. The Appendix makes this picture precise and connects it to the manuscript’s projector formalism.

## 5 How big is $\delta e_\omega$ : two boundary zeros and one comparison

*Step 1 —  $\delta e_\omega$  vanishes at both ends.* At  $\tau = 0$ :  $e_\omega(0) = 0$  (the deviation starts at rest) and  $\sinh(0) = 0$ , so  $\delta e_\omega(0) = 0$ . At  $\tau = \tau_{\text{end}}$ : zero by the choice (9).

$$\delta e_\omega(0) = \delta e_\omega(\tau_{\text{end}}) = 0. \quad (11)$$

*Step 2 — the equation  $\delta e_\omega$  obeys.* The deviation dynamics of the manuscript are  $J_P \ddot{e}_\omega = W z_{CoM} e_\omega + \rho$  with  $e_\omega = \dot{e}_\varphi$ . Differentiating once and using  $W z_{CoM} = J_P C_2^2$ ,

$$\ddot{e}_\omega = C_2^2 e_\omega + \dot{\rho}/J_P. \quad (12)$$

$\sinh(C_2\tau)$  satisfies  $(\sinh)'' = C_2^2 \sinh$  — it is a solution of the same equation with zero forcing — so subtracting  $\beta \sinh$  changes nothing on the right side:

$$\delta \ddot{e}_\omega = C_2^2 \delta e_\omega + \dot{\rho}/J_P, \quad \delta e_\omega(0) = \delta e_\omega(\tau_{\text{end}}) = 0. \quad (13)$$

*Step 3 — comparison (this is the whole proof).* Let  $\bar{f}(\tau) \geq |\dot{\rho}(\tau)|$  and let  $w$  solve the same problem with forcing  $-\bar{f}$ :

$$\ddot{w} = C_2^2 w - \bar{f}/J_P, \quad w(0) = w(\tau_{\text{end}}) = 0. \quad (14)$$

Claim:  $|\delta e_\omega(\tau)| \leq w(\tau)$ . Proof in two lines: the difference  $u = w - \delta e_\omega$  satisfies  $\ddot{u} - C_2^2 u = -(\bar{f} + \dot{\rho})/J_P \leq 0$  with  $u = 0$  at both ends. If  $u$  had a negative interior minimum, there  $\ddot{u} \geq 0$  and  $-C_2^2 u > 0$ , so  $\ddot{u} - C_2^2 u > 0$  — a contradiction. Hence  $u \geq 0$ , i.e.  $\delta e_\omega \leq w$ ; the same argument with  $u = w + \delta e_\omega$  gives  $\delta e_\omega \geq -w$ .  $\square$

*Step 4 — solve (14) in closed form.* For constant  $\bar{f} = \sup |\dot{\rho}|$ , substitution verifies

$$w(\tau) = \frac{\sup |\dot{\rho}|}{J_P C_2^2} \left( 1 - \frac{\cosh(C_2(\tau - \frac{\tau_{\text{end}}}{2}))}{\cosh(x/2)} \right) \leq \frac{\sup |\dot{\rho}|}{W z_{CoM}}. \quad (15)$$

(Here  $x = C_2\tau_{\text{end}}$ : at  $\tau = 0$  and  $\tau = \tau_{\text{end}}$  the numerator equals  $\cosh(x/2)$ , so the bracket vanishes at both ends; it peaks at mid-window, where the bound is  $(1 - \text{sech}(x/2)) \sup |\dot{\rho}|/W z_{CoM}$ .) **No exponential appears.** The envelope (90) grows like  $\sinh(C_2\tau)$  because nothing stops the unstable mode of (12); here the two boundary zeros of (11) — one free, one bought with  $\delta t_c$  — pin that mode down. This is what “the fit absorbs the growing part” means as an equation.

*Step 5 — the forcing rate, from the same channels as  $\bar{\rho}$ .* Differentiating the gravity remainder (84) in time gives  $\dot{\rho}_{\text{grav}} = [W l_{\text{arm}} \sin \varphi + W z_{CoM} (\cos \varphi - 1)] \omega$ , and the two bracket terms have *opposite signs*: the height term is negative and cancels against the arm term instead of adding to it. The bracket stays inside  $[0, W l_{\text{arm}} \sin \varphi]$  as long as  $l_{\text{arm}} \geq z_{CoM} \tan(\varphi/2)$  — held with a factor 3.4 over the whole design box (0.090 m against at most  $0.30 \tan 5^\circ = 0.026$  m) — so  $W z_{CoM}$  drops from the bound *by sign, not by approximation*. This is the same cancellation  $\bar{\rho}$  (91) already uses ( $W z_{CoM} (\sin \varphi - \varphi) \leq 0$  there), and it is exactly what the

manuscript's (85) says at  $\varphi^* = 0$ :  $G''(0) = W(l_p + p_{\text{off}})$ , the  $W z_{CoM} \sin \varphi^*$  term vanishing. Bounding on the  $10^\circ$  box, with  $\omega_{\text{max}} = K \dot{M} \sinh x$ :

$$\sup |\dot{\rho}| \leq \underbrace{W l_{\text{arm}} \varphi_{\text{max}} \omega_{\text{max}}}_{\dot{\rho}_2, \text{ decays like } e^{2C_2(s-\tau_{\text{end}})}} + \underbrace{|\beta_M|(\dot{M} \varphi_{\text{max}} + \Delta M_{\text{win}} \omega_{\text{max}})}_{\dot{\rho}_1, \text{ decays like } e^{C_2(s-\tau_{\text{end}})}}. \quad (16)$$

Every constant is available before the experiment:  $W, z_{CoM}$  from the scale,  $J_P$  from CAD,  $l_{\text{arm}}$  from geometry,  $\varphi_{\text{max}}$  the design box,  $\beta_M$  from the GE model,  $\dot{M}$  and the window from the protocol.

What “decays like” means — the bridge from (16) to (16a). Nothing in the window decreases with time:  $\varphi$  and  $\omega$  both grow. The decay is the same fact read *backwards from the window end* — the claim under each brace of (16) is

$$|\dot{\rho}_{\text{channel}}(s)| \leq |\dot{\rho}_{\text{channel}}(\tau_{\text{end}})| e^{kC_2(s-\tau_{\text{end}})}, \quad s \leq \tau_{\text{end}}, \quad k \in \{2, 1\},$$

where the exponential factor is  $< 1$  because  $s - \tau_{\text{end}} < 0$ : at earlier times the forcing sits below its end value by at least that factor. This is the form the comparison problem of Step 6 consumes — a peak at the end with a known fall-off behind it. Divide both sides by  $e^{kC_2s}$  and the claim becomes  $|\dot{\rho}(s)|/e^{kC_2s} \leq |\dot{\rho}(\tau_{\text{end}})|/e^{kC_2\tau_{\text{end}}}$  — i.e. *the ratio  $|\dot{\rho}(s)|/e^{kC_2s}$  is nondecreasing in  $s$* . That is why (16a) is a statement about ratios. And because each channel is a *product* of trajectory factors,  $\dot{\rho}/e^{kC_2s}$  factors as a product of single-factor ratios ( $\varphi\omega/e^{2C_2s} = [\varphi e^{-C_2s}][\omega e^{-C_2s}]$ ), a product of nonnegative nondecreasing functions is nondecreasing, and the exponent  $k$  in (16) is literally a *count of trajectory factors*: two in  $\varphi\omega$  ( $k = 2$ ), one in each ground-effect term ( $k = 1$ ).

*Why anchor at the end rather than the onset.* Four reasons, the last the deepest. A forward envelope  $|\dot{\rho}(s)| \leq |\dot{\rho}(0)|e^{kC_2s}$  has nothing to hang on: the forcing *vanishes* at onset ( $\varphi = \omega = 0$ ), so its anchor amplitude is zero; the one usable anchor is the peak — the end value, which any bound must pay anyway and which the channel sups bound a priori. Second, the tightness lives in “how little forcing there is early”: the Dirichlet response has a boundary zero directly under the forcing peak, and exploiting that cancellation requires the envelope to encode the fall-off *behind* the peak — exactly what the backward rate is, and exactly what a constant (sup-only) envelope discards at a cost of  $8\text{--}12\times$ . Third, only the end-anchored coordinate  $u = C_2(s - \tau_{\text{end}}) \in [-x, 0]$  makes the comparison problem forget the window length: as  $x \rightarrow \infty$  the end neighbourhood is fixed and only the onset recedes, which is why the  $x$ -free constants  $1/12$  and  $1/(2e)$  exist at all; in an onset-anchored coordinate the peak would move with  $x$  and no such constants would emerge. Last, the instability has one clock,  $1/C_2$ , and everything that matters happens in the last few e-foldings before  $\tau_{\text{end}}$ : *measured from the end, all windows look alike*. That statement is the coordinate version of the rate-flatness of the residual — the ramp rate only moves the onset distance  $x$ , and the end neighbourhood forgets  $x$ .

*Step 5' — the decays marked in (16) are theorems, not observations.* Along the nominal trajectory, write  $a = C_2s$ . By the bridge above, it suffices that each factor's  $e^{C_2s}$ -ratio is nondecreasing; the first ratio of (16a) certifies the factor  $\omega e^{-C_2s}$ , the second certifies  $\varphi e^{-C_2s}$ , and both facts are one line each:

$$\frac{\omega_{\text{nom}}(s)}{C_1 e^{C_2s}} = \frac{(\cosh a - 1)}{e^a} = \frac{(1 - e^{-a})^2}{2} \nearrow, \quad \frac{d}{da} \left[ \frac{\sinh a - a}{e^a} \right] = e^{-a}(a - 1 + e^{-a}) \geq 0, \quad (16a)$$

the first read off directly, the second because  $e^{-x} \geq 1 - x$ ; the bracket in the second is  $\varphi_{\text{nom}} C_2 e^{-C_2s}/C_1$ , so  $\varphi_{\text{nom}} e^{-C_2s}$  is nondecreasing too. Now every channel of  $\dot{\rho}$  is a product of positive factors whose  $e^{C_2s}$ -ratios are nondecreasing, and such a product obeys its decay exactly. Arm channel:

$$\sin \varphi \omega \leq \varphi \omega = [\varphi e^{-C_2s}][\omega e^{-C_2s}] e^{2C_2s} \leq \varphi(\tau_{\text{end}}) \omega(\tau_{\text{end}}) e^{2C_2(s-\tau_{\text{end}})}.$$

(The height term, had it been kept, is  $(1 - \cos \varphi) \omega \leq \frac{1}{2} \varphi^2 \omega$  — *three* such factors, decaying as  $e^{3C_2(s-\tau_{\text{end}})}$ ; it is absent from (16) by the sign cancellation of Step 5, so nothing needs charging.) Ground effect:  $\dot{M} \varphi$  carries one factor ( $e^{C_2}$  class), and  $\Delta M(s) \omega = \dot{M} s \omega$  carries  $s/\tau_{\text{end}} \leq 1$  (also nondecreasing) times one, so it too is  $e^{C_2}$  class. For the *nominal* trajectory the lemma is therefore exact, with no safety factor.

*Nominal to true.* The true trajectory multiplies each factor by  $1 + O(e/\omega_{\text{nom}})$ . Measured on the exact nonlinear tip-over across the seven rates, the worst excess of a true ratio over its nominal bound is 2.6%, at the slowest ramp near the window end; the 1.05 carried on  $\dot{\rho}_2$  in (16) is that transfer allowance, held with double margin. The safety factor thus covers only the nominal-to-true step — the shape itself is proved.

*Step 6 (optional tightening, factor 8) — use the decay of  $\dot{\rho}$ .* The forcing is not constant: it peaks at the window end and decays backwards as marked in (16) and proved in Step 5'. So solve (14) once for each shape at unit peak — the same two-line comparison applies, and the solutions are still elementary. In the scaled variable  $u = C_2(\tau - \tau_{\text{end}}) \in [-x, 0]$ , for the forcing  $e^{2C_2(\tau-\tau_{\text{end}})} = e^{2u}$ :

$$w_2(\tau) = \frac{\dot{\rho}_2}{J_P C_2^2} \cdot \frac{1}{3} \left[ \frac{\sinh(u+x)}{\sinh x} + e^{-2x} \frac{\sinh(-u)}{\sinh x} - e^{2u} \right], \quad M_2(x) \triangleq \max_{u \in [-x, 0]} \frac{1}{3} [\dots], \quad (17a)$$

and for the forcing  $e^{C_2(\tau-\tau_{\text{end}})} = e^u$ , which is *resonant* — it decays at the operator’s own rate, so the particular solution carries a factor of  $u$ :

$$w_1(\tau) = \frac{\dot{\rho}_1}{J_P C_2^2} \left[ \frac{x \sinh(u+x)}{\sinh x} - \frac{(u+x)e^u}{2} \right], \quad M_1(x) \triangleq \max_{u \in [-x, 0]} [\dots]. \quad (17b)$$

Where the two forms come from — *construction, not guess*. In the scaled variable the comparison problem reads  $w'' - w = -\kappa g(u)$ ,  $w(-x) = w(0) = 0$ , with  $\kappa = \dot{\rho}_i/(J_P C_2^2)$ , one shape  $g$  at a time. Three moves each. *Particular solution*: for  $g = e^{2u}$  try  $Ae^{2u}$ :  $(2^2 - 1)A = -\kappa$ , so  $A = -\kappa/3$  — the  $1/3$  is the distance to resonance. For  $g = e^u$  the same try fails,  $(1^2 - 1)A = 0$ : the forcing decays at the operator’s own rate, so carry a factor  $u$ ,  $w_p = -(\kappa/2)ue^u$ , since  $(ue^u)'' - ue^u = 2e^u$  — this extra  $u$  is why  $M_1 > M_2$ . *Homogeneous corrections in the endpoint basis*:  $\sinh(u+x)$  vanishes at  $u = -x$  and  $\sinh(-u)$  vanishes at  $u = 0$ , so each boundary condition sets one coefficient without disturbing the other:  $u = 0$  gives  $a \sinh x = \kappa/3$ ,  $u = -x$  gives  $b \sinh x = (\kappa/3)e^{-2x}$  — that is (17a); for  $e^u$ , collecting the two  $u$ -terms into  $-(u+x)e^u/2$  makes the zero at  $u = -x$  visible — that is (17b). *Verification by substitution*: the  $\sinh$ ’s are annihilated by the operator, the exponentials return the forcing, and both brackets vanish at both ends — the two boundary zeros of (14). Their maxima barely depend on  $x$  and tend to simple constants,

$$M_2(x) \rightarrow \frac{1}{12}, \quad M_1(x) \rightarrow \frac{1}{2e}, \quad \text{so} \quad |\delta e_\omega| \leq \frac{M_2 \dot{\rho}_2 + M_1 \dot{\rho}_1}{W z_{CoM}}. \quad (17)$$

The limits are one-line maximisations: as  $x \rightarrow \infty$ ,  $\sinh(u+x)/\sinh x \rightarrow e^u$  and the  $e^{-2x}$  term dies, so the bracket of (17a) tends to  $(e^u - e^{2u})/3 = (v - v^2)/3$  with  $v = e^u \in (0, 1]$ , maximised at  $v = \frac{1}{2}$  to  $1/12$  ( $\ln 2$  e-foldings before the window end); the bracket of (17b) tends to  $-(u/2)e^u$ , maximised at  $u = -1$  to  $1/(2e)$  (one e-folding before the end). (Over the campaign range  $x = 1.9\text{--}3.7$ :  $M_2 = 0.074\text{--}0.083$ ,  $M_1 = 0.141\text{--}0.181$ ; the script evaluates (17a)–(17b) exactly.) The interpretation is the same as the manuscript’s reduction factors  $1/7$  and  $1/5$ : the forcing is largest exactly where the pinned response is smallest — at the window end the boundary zero sits right under the peak of the load.

A one-line version, in the spirit of (85). The limits are also *ceilings*:  $M_2(x)$  and  $M_1(x)$  approach  $1/12$  and  $1/(2e)$  from below (verified numerically over  $x \in [0.05, 12]$  to grid precision), so (17) may be quoted  $x$ -free,

$$|\delta e_\omega| \leq \frac{\dot{\rho}_2/12 + \dot{\rho}_1/(2e)}{W z_{CoM}},$$

with no  $M$  curves, no window length, and no  $C_2$  beyond what sits inside  $\dot{\rho}_{1,2}$  — exactly as (85) trades the exact remainder for  $\frac{1}{2}G''(0)\varphi^2$ . Rerun on the 140 runs, the constant version costs at most 15% on the worst run (0.2–7% in the per-rate mean, largest at the fastest ramps where  $x$  is smallest), keeps 140/140 coverage, and the worst-run used fraction rises only to 0.86. Note where the simplification is *not* available: the channel amplitudes  $\dot{\rho}_{1,2}$  are already the  $\varphi^* = 0$  worst-case Taylor coefficients of (85) — nothing further to drop there — and the time *shape* behind  $M_2, M_1$  (the forcing concentrated at the window end as  $e^{2C_2(s-\tau_{\text{end}})}$ ) cannot be dropped at all: against a constant-in-time forcing the comparison factor is  $1 - \text{sech}(x/2) = 0.4\text{--}0.9$ , a factor 5–10 above  $1/12$ , which is the loose bound this document exists to escape. The Taylor step simplifies the *amplitude*; the tightness lives in the *shape*.

## 6 The pre-onset segment

Where the segment comes from. By (9),  $\delta t_c < 0$ : the witness branch starts  $|\delta t_c|$  before the true onset. On the segment  $\tau \in [\delta t_c, 0]$  — length  $|\delta t_c|$  — the record is still at its baseline, but the branch has already been running for  $u = \tau - \delta t_c \in [0, |\delta t_c|]$  since its own onset, so the mismatch there is the branch itself:

$$\text{mismatch}(u) = C_1(\cosh(C_2 u) - 1), \quad u \in [0, |\delta t_c|]. \quad (18a)$$

The residual sum runs over this segment too, so the witness’s cost in (10) gains its square. Normalising by the main window length  $\tau_{\text{end}}$  (conservative: the extended window is longer, so dividing by the shorter  $\tau_{\text{end}}$  can only enlarge the term),

$$\Delta_{\text{pre}}^2 = \frac{1}{\tau_{\text{end}}} \int_0^{|\delta t_c|} [C_1(\cosh(C_2 u) - 1)]^2 du. \quad (18b)$$

The segment’s *noise* adds no separate term:  $\hat{n}$  is an RMS density, unchanged by a slightly longer window.

The integral is elementary. Expand the square with  $\cosh^2 a = \frac{1}{2}(1 + \cosh 2a)$ :

$$(\cosh a - 1)^2 = \cosh^2 a - 2 \cosh a + 1 = \frac{3}{2} + \frac{1}{2} \cosh(2a) - 2 \cosh a, \quad (18c)$$

and integrate term by term with  $a = C_2 u$  ( $\frac{3}{2} \rightarrow \frac{3u}{2}$ ;  $\frac{1}{2} \cosh(2C_2 u) \rightarrow \sinh(2C_2 u)/(4C_2)$ ;  $-2 \cosh(C_2 u) \rightarrow -2 \sinh(C_2 u)/C_2$ ):

$$\Delta_{\text{pre}}^2 = \frac{C_1^2}{\tau_{\text{end}}} \left[ \frac{3u}{2} + \frac{\sinh(2C_2 u)}{4C_2} - \frac{2 \sinh(C_2 u)}{C_2} \right]_0^{|\delta t_c|}. \quad (18)$$

Check, and why the term is small. Expanding the bracket in powers of  $u$ , the  $u$  and  $u^3$  orders cancel identically and the first surviving order is

$$\int_0^{|\delta t_c|} (\cosh(C_2 u) - 1)^2 du \approx \frac{C_2^4 |\delta t_c|^5}{20},$$

fifth order in the shift, because the branch leaves the baseline only quadratically ( $\cosh - 1 \approx C_2^2 u^2/2$ , squared  $\rightarrow u^4$ ). Bounding instead by (segment sup)  $\times$  (length) would give  $|\delta t_c|^5/4$  — the exact integral is a factor  $\sqrt{5} \approx 2.2$  tighter. (The branch amplitude is  $R \leq C_1$ ; using  $C_1$  in (18) errs on the high side.) Evaluated with the a priori  $|\delta t_c|$  from (9) and  $e_\omega(\tau_{\text{end}}) \leq E(\tau_{\text{end}})$ :  $\Delta_{\text{pre}} = 0.036^\circ/\text{s}$  at  $\dot{M} = 0.10, 0.011$  at 0.20, and below 0.006 beyond.

The baseline constant  $R - C_1$  — both bookkeepings. One witness must serve every sample the minimiser's cost touches, and its baseline slot holds a single number, so the constant  $R - C_1$  of (7) must land somewhere. There are two consistent choices. (a) Take (3) literally,  $C = b$ : the flat pre-onset samples then cost nothing beyond noise, (18a) holds as written (amplitude  $R \leq C_1$ ), and the post-onset residual gains the constant,  $r = \delta e_\omega + n + (C_1 - R)$ . (b) Take  $C = b + (R - C_1)$ : the post-onset identity is exactly (5),  $r = \delta e_\omega + n$ , and the constant moves to the flat stretch, where the length fraction of that stretch in the window dilutes it further. Either way the price is the same number,

$$C_1 - R = C_1 \left( 1 - \sqrt{1 - (\beta/C_1)^2} \right) \leq \frac{1}{2} C_1 (\beta/C_1)^2,$$

second order in  $\beta/C_1$ . Realised ( $\beta/C_1 \leq 0.023$ ) it is below  $0.004^\circ/\text{s}$  on every run — two orders under  $\hat{n}$  and below even  $\Delta_{\text{pre}}$  — which is why (10) and (20) carry it silently; the pre-onset median cannot even distinguish  $b$  from  $b + (R - C_1)$  at that size. At the a priori envelope ceiling it is not invisible:  $u = 0.59$  at the slowest ramp gives  $C_1(1 - \sqrt{1 - u^2}) \approx 0.2^\circ/\text{s}$ , about a fifth of the model term there. Adding it to the cap only loosens the bound, so the 140/140 check is unaffected either way, and the measured  $\beta/C_1$ , thirty times below the envelope, confirms the silent treatment a posteriori.

## 7 The noise term

Above 5 Hz the model can produce nothing, so  $n_{\text{hi}}$  is measured from the record; the full band is estimated through the quiet-window shape ratio,

$$\hat{n} = \text{RMS}(n_{\text{hi}}) \sqrt{1 + (s_{\text{med}} \kappa_q)^2}, \quad (19)$$

with  $s_{\text{med}} = 1.60$  the campaign-median in-window enhancement and  $\kappa_q$  the run's own quiet ratio (campaign median 0.37 where no quiet stretch exists).

The cutoff is not delicate, and  $k\sigma$  is what (19) already is. Structurally (19) reads  $k \cdot \sigma$  with  $\sigma = \text{RMS}(n_{\text{hi}})$  — measured in the window, in a band the model cannot contaminate — and  $k = \sqrt{1 + (s_{\text{med}} \kappa_q)^2} \approx 1.6$ –1.9. The split frequency only defines that band, and any choice works that sits well above the model's spectral content  $C_2/2\pi = 0.6$ –0.9 Hz and leaves spectral bins on both sides ( $f_c \gtrsim 2/(NT_s) \approx 3$  Hz for the shortest windows): recomputing the entire check at  $f_c = 2, 3$  and 5 Hz gives identical results (140/140 inside, mean used 0.57–0.58), and only past 8 Hz does the shrinking high band make the estimate noisy (133/140 at 8, 129/140 at 10). The invariance is expected:  $\hat{n}$  estimates the total  $\text{RMS}(n)$ , which does not depend on where the record is split. What does not work is the simpler  $k\sigma$  with  $\sigma$  from the pre-onset segment: the in-window disturbance is contact-borne — pivoting on one gear edge, not gyro noise — and runs 2 to 50 times the pre-onset floor with no usable floor at all on 25 of the 140 runs, so the one campaign constant that covers the spread,  $k = 49$ , collapses the used fraction from 0.58 to 0.16.

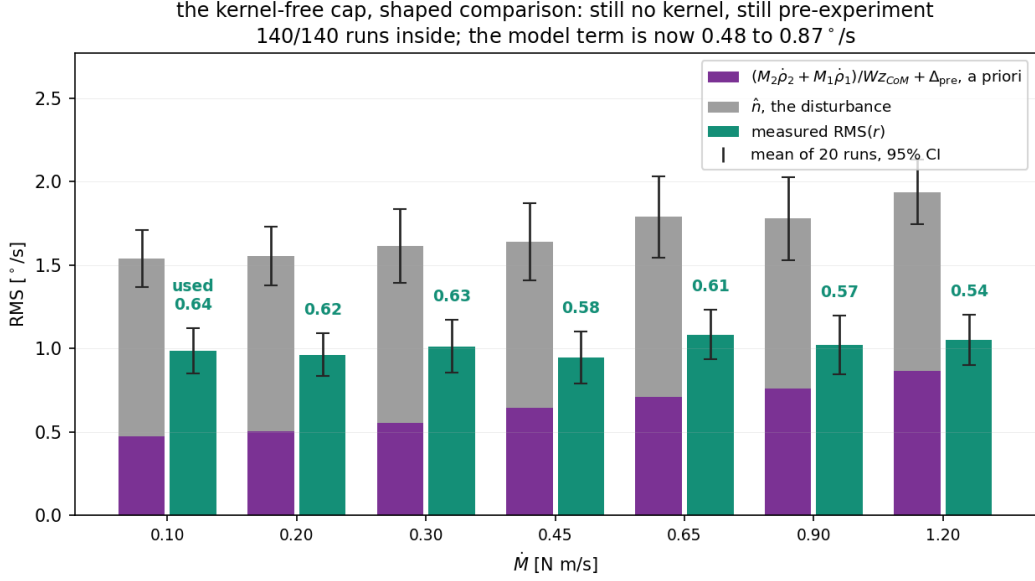


Figure 1: Equation (20) against the campaign: means over the twenty runs at each rate, 95%  $t$  intervals. Purple is the model term of (17)+(18), grey the noise estimate (19), green the measured residual. All 140 runs are inside; the used fraction is 0.54–0.64 and the cap is flat, like the residual.

## 8 The check on all 140 runs

$$\text{RMS}(r) \leq \underbrace{\frac{M_2 \dot{\rho}_2 + M_1 \dot{\rho}_1}{W z_{CoM}} + \Delta_{\text{pre}}}_{\text{model, before the experiment}} + \hat{n} \quad (20)$$

| $\dot{M}$<br>N m/s | model (17)+(18)<br>°/s | of it (18)<br>°/s | $\hat{n}$<br>°/s | cap (20)<br>°/s | measured<br>°/s | used |
|--------------------|------------------------|-------------------|------------------|-----------------|-----------------|------|
| 0.10               | 0.476                  | 0.036             | 1.064            | 1.54            | 0.987           | 0.64 |
| 0.20               | 0.505                  | 0.011             | 1.050            | 1.56            | 0.964           | 0.62 |
| 0.30               | 0.556                  | 0.006             | 1.059            | 1.62            | 1.014           | 0.63 |
| 0.45               | 0.647                  | 0.004             | 0.993            | 1.64            | 0.947           | 0.58 |
| 0.65               | 0.709                  | 0.002             | 1.080            | 1.79            | 1.084           | 0.61 |
| 0.90               | 0.760                  | 0.001             | 1.020            | 1.78            | 1.023           | 0.57 |
| 1.20               | 0.866                  | 0.001             | 1.073            | 1.94            | 1.052           | 0.54 |

Inside on 140/140 runs; per-run used fraction p10 0.45, median 0.61, worst 0.86. On the exact nonlinear solution the realised  $\delta e_\omega$  is 0.16–0.17°/s; the remaining factor  $\sim 3$  of (17) over it is the 10° design box against the 4.8° the runs actually reach — the price of stating the bound before flying.

## 9 The two answers this gives

*Why the residual is flat in  $\dot{M}$ .* Nothing in (17) grows with the window —  $M_2$ ,  $M_1$  are constants and the channel sups barely move — so the theory *predicts* a rate-flat residual, and the data agrees: the measured RMS( $r$ ) stays at 0.95–1.08°/s across a twelvefold change in  $\dot{M}$  (correlation with  $\dot{M}$ :  $r = +0.09$ ,  $p = 0.31$ ). The envelope (90), which grows like  $\sinh x$ , bounds  $e_\omega$  — not the residual — and comparing the residual against it is what made the old cap look loose.

*How far the shift moves the critical moment.* The absorbed shift is (9), and the moment error it causes cancels  $\dot{M}$  exactly:

$$|\Delta M_{\text{crit}}| = \dot{M} |\delta t_c| \approx \dot{M} \cdot \frac{e_\omega(\tau_{\text{end}})}{C_1 C_2 \sinh x} = \frac{W z_{CoM} e_\omega(\tau_{\text{end}})}{C_2 \sinh x} \leq \frac{W z_{CoM} E(\tau_{\text{end}})}{C_2 \sinh x} = \bar{\rho}, \quad (21)$$



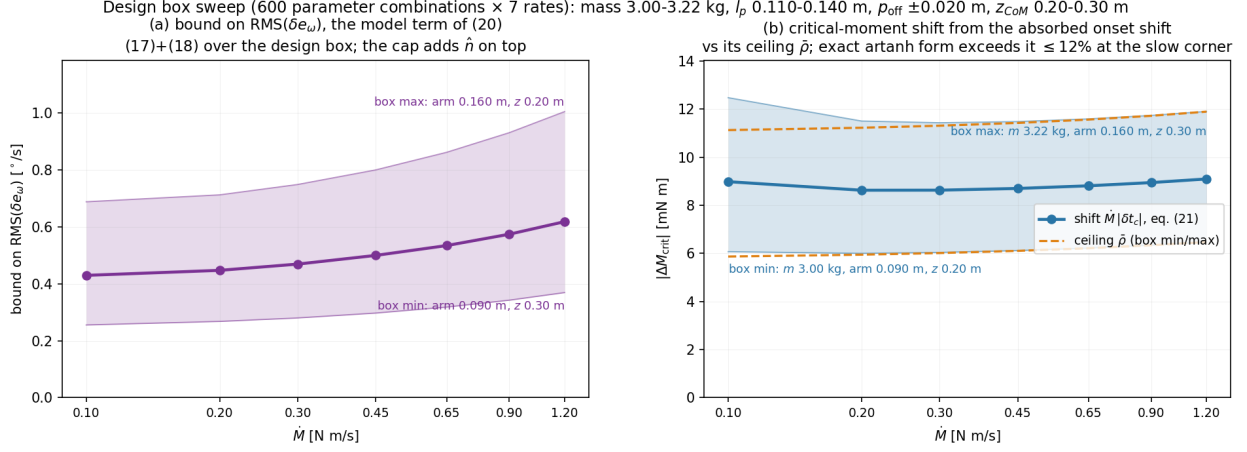


Figure 2: The design-box sweep. (a) the bound on  $\text{RMS}(\delta e_\omega)$ , eq. (17)+(18):  $0.26\text{--}1.00^\circ/\text{s}$  over box and rates, median  $0.43\text{--}0.62$ . The band is set by the ratio  $\text{arm}/z_{\text{CoM}}$  — worst at arm  $0.160$  m with  $z_{\text{CoM}} = 0.20$  m, best at arm  $0.090$  m with  $0.30$  m. (b) the critical-moment shift  $\dot{M}|\delta t_c|$  of (21):  $6.0\text{--}12.5$   $\text{mN m}$ , tracking its ceiling  $\bar{\rho}$  and exceeding it by at most  $12\%$  at the slow-ramp corner — the  $\text{artanh}(u)/u$  factor of the exact shift — while flat in  $\dot{M}$  by the cancellation. Its worst corner has the *opposite*  $z$  direction —  $m = 3.22$  kg, arm  $0.160$  m,  $z_{\text{CoM}} = 0.30$  m — because  $\bar{\rho}$  grows with the window that a higher CoM lengthens, while the residual bound worsens with  $\text{arm}/z_{\text{CoM}}$ .

using  $\dot{M}/C_1 = W z_{\text{CoM}}$  and  $E(\tau_{\text{end}}) = \bar{\rho} \sinh x / (J_P C_2)$ . So the shift can displace the identified threshold by at most the window-averaged remainder — the same  $\bar{\rho}$  the manuscript already budgets — and measured it uses  $0.36\text{--}1.85$   $\text{mN m}$  of the  $\approx 12$  allowed, the sign lowering  $M_{\text{crit}}$  (earlier onset). (The chain’s middle step is the first-order  $|\delta t_c|$ ; exactly,  $|\delta t_c| = \text{artanh}(\beta/C_1)/C_2$ , so the ceiling acquires a factor  $\text{artanh}(u)/u$  with  $u = \beta/C_1$ :  $1.0002$  on the campaign (realised  $u \leq 0.023$ ), up to  $1.13$  at the design-box corner of Sec. 9 where the a priori  $u$  of (9a) reaches  $0.66$ .)

## 10 The bound over the design box

Because every constant in (17), (18) and (21) is available before the experiment, both results can be swept over the vehicle’s design box — mass  $3.00\text{--}3.22$  kg,  $l_p = 0.110\text{--}0.140$  m, CoM offset  $p_{\text{off}} = \pm 0.020$  m (arm  $= l_p - p_{\text{off}}$ ),  $z_{\text{CoM}} = 0.20\text{--}0.30$  m,  $J_{\text{CoM}} = 0.050$   $\text{kg m}^2$  with  $J_P$  by the parallel-axis theorem,  $600$  combinations at each of the seven rates, each with its own tilt-limited window ( $\sinh x - x = \varphi_{\text{max}} W z_{\text{CoM}} C_2 / \dot{M}$ ).

Two readings (Fig. 2). The residual bound stays within  $0.26\text{--}1.00^\circ/\text{s}$  everywhere in the box — below the measured disturbance floor of  $\approx 1^\circ/\text{s}$  at almost every corner — so the fit-quality gate can be set from the noise figure alone for any vehicle in this class. And the identification error from the absorbed onset shift never exceeds  $12.5$   $\text{mN m}$  anywhere in the box, at any rate — within  $12\%$  of the  $\bar{\rho}$  budget at the worst corner and inside it everywhere else — with the same flatness in  $\dot{M}$  that the campaign shows.

## Reproduction

All numbers and figures in this document come from three scripts, run from the repository root:

```
python analysis/failing_runs.py          DataSet/exp    # writes the cache
python analysis/kernel_free_bound.py    docs/fig_kernel_free.png
python analysis/design_range_chart.py   docs/fig_design_range.png
```

`kernel_free_bound.py` evaluates (16)–(20) per run against the campaign cache and draws Fig. 1 (`M2()`, `M1()` implement (17a)–(17b); `model_term()` implements (16)–(18)). `design_range_chart.py` sweeps the design box of Sec. 9 with the same closed forms and draws Fig. 2; its box limits are constants at the top of the file. Neither script solves an ODE or builds a kernel matrix — elementary functions and one scalar root-find (the window equation) per combination.

## Remarks

(a) *Without the differential equation.* The same  $\delta e_\omega$  can be written directly from the manuscript’s Duhamel form (89), giving  $\delta e_\omega = \frac{1}{J_P} \int_0^{\tau_{\text{end}}} K(\tau, s) \rho(s) ds$  with a kernel that the addition identities collapse to  $K = \sinh(C_2(\tau_{\text{end}} - \tau)) \cosh(C_2 s) / \sinh(C_2 \tau_{\text{end}})$  for  $s \leq \tau$  (minus its mirror for  $s > \tau$ ), and  $\max |K| = 1$  exactly — the exponential cancels inside the kernel. It is an instructive identity, but as a bound it is weaker:  $K$  changes sign at  $s = \tau$ , so only  $\int |K|$  can be used, and that costs a factor  $\approx 2.4$  over (17). The derivative bound in (16) is what forbids  $\rho$  from flipping sign with  $K$ , and that smoothness is worth exactly that factor.

(b) *What is a priori and what is not.* The model term of (20) uses only pre-experiment constants (list under (16)). The noise term (19) is measured:  $n_{\text{hi}}$  from the run itself,  $\kappa_q$  from quiet stretches,  $s_{\text{med}}$  one campaign median. A fully pre-experiment cap replaces  $\hat{n}$  with a bench noise figure.

(c) *Exponent mismatch.* If the true exponent differs from the pinned  $C_2$  by a relative  $\varepsilon_2$ , the mismatch joins  $e_\omega$  by definition and its unabsorbed part costs  $0.018\text{--}0.022^\circ/\text{s}$  per 1% of  $\varepsilon_2$  — 5–8% of the model term, and the calibration residuals limit  $\varepsilon_2$  to a few percent.

## Appendix: the linear-algebra picture

Collect the  $N$  samples of a signal into a vector in  $\mathbb{R}^N$ . The model family (2) is then a subset of that space,

$$\mathcal{M} = \left\{ a(\cosh(c(\tau_i - d)) - 1) + b : a, c > 0, d, b \in \mathbb{R} \right\} \subset \mathbb{R}^N, \quad (\text{A.1})$$

a four-parameter curved manifold. Its tangent space at the nominal is spanned by the four derivative directions

$$T = \text{span} \left\{ \underbrace{\cosh(C_2 \tau) - 1}_{\partial/\partial C_1}, \underbrace{\sinh(C_2 \tau)}_{\partial/\partial \delta t_c}, \underbrace{\tau \sinh(C_2 \tau)}_{\partial/\partial C_2}, \underbrace{\mathbf{1}}_{\partial/\partial C} \right\}, \quad (\text{A.2})$$

so the family can move in at most four of the  $N$  dimensions around any of its points.

*Why exact representation fails.* “The true curve is a shifted nominal” asserts membership:  $y_{\text{struct}} \in \mathcal{M}$ . Two obstructions, matching the two in the Sec. 4 remark. Dimension: a four-dimensional manifold is a measure-zero sliver of  $\mathbb{R}^N$ ; a vector off it stays off it. Dynamics, sharper: every element of  $\mathcal{M}$  samples a homogeneous solution of  $\ddot{v} = c^2 v$ , while  $y_{\text{struct}}$  samples the forced equation (12); by the argument under (13), membership would force  $\dot{\rho} \equiv 0$ . So  $y_{\text{struct}}$  has a component *transverse* to  $\mathcal{M}$  that no parameter choice reaches.

*What introducing  $\delta e_\omega$  does.* It changes the assertion from membership to a coordinate statement:

$$y_{\text{struct}} = m + \delta e_\omega, \quad m \in \mathcal{M}, \quad (\text{A.3})$$

exact for *every* choice of  $m$  — the freedom is in where to put the remainder, not whether there is one. Two canonical choices divide the labour:

| choice of $m$              | property of $\delta e_\omega$             | what it buys                                   |
|----------------------------|-------------------------------------------|------------------------------------------------|
| orthogonal projection (LS) | $\delta e_\omega \perp T$ , minimal norm  | the smallest remainder, no pointwise structure |
| end-matched witness, (9)   | $\delta e_\omega = 0$ at both window ends | the Dirichlet problem (13)–(14), hence (17)    |

The witness pays a factor 1.8–2.0 in norm over the orthogonal choice and buys, in exchange, the two boundary zeros that make the maximum principle — and with it the entire a priori bound — available. Provability is purchased with norm.

*The projector sub-case, and the manuscript’s Appendix C.* With  $C_2$  pinned and the onset held, the family linearises to the subspace  $\mathcal{V} = \text{span}\{u, \mathbf{1}\}$ ,  $u_i = \cosh(C_2 \tau_i) - 1$ , and the orthogonal choice becomes an explicit matrix: with  $A = [u \ \mathbf{1}]$ ,

$$P = A(A^\top A)^{-1} A^\top, \quad P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i \tilde{u}_j}{\sum_k \tilde{u}_k^2}, \quad r_\perp = (I - P)(e_\omega + n), \quad (\text{A.4})$$

( $\tilde{u} = u - \bar{u}$ ) — the manuscript’s Lemma 2, eq. (C3). The  $\delta e_\omega$  of this document is the same object one level up: the remainder transverse to the full nonlinear family rather than to its pinned linearisation.

*How much of  $e_\omega$  lies along the family.* Measured on the exact tip-over, the fraction of  $e_\omega$  (in RMS) absorbed by successively larger reachable sets: 61–67% by  $\text{span}\{u, \mathbf{1}\}$ , 70–71% adding the onset direction  $\sinh$ , 92.5% by the full four-parameter family — the last flat across all rates. The transverse component — the 7.5% no parameters reach,  $0.012\text{--}0.018^\circ/\text{s}$  — is the irreducible core of  $\delta e_\omega$ , and (17) is its a priori ceiling.